

MUGE 38-System Extended Environment: F_torque + F_shock + F_cosmo Auto-Cascade

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PAPER_457 — MUGE 38-System Extended Environment: F_torque + F_shock + F_cosmo Auto-Cascade

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Classification: FIRST F_torque gravitational tidal torque term in UQFF; FIRST F_shock supernova/stellar-wind shock front term; FIRST auto-cascade from QG+DM+GW to F_cosmo

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CP4 Class: MUGECompressed38SystemExtendedEnvCalculator (#95, PAPER_457)

<!-- UQFF constants: = 5.0e-4 day-1, [SSq] = 0.57, F_torque = 1×10-11 m/s2, F_shock = 1×10-11 m/s2 --> —

Abstract

The 38-system MUGE extension expands the Session 116 29-system registry by adding 9 new object classes including the Lagoon Nebula, spiral supernovae host NGC-class systems, bipolar planetary nebula NGC 6302, and the full Orion region — while introducing two new fundamental environmental factor types: F_torque (gravitational tidal torque) and F_shock (supernova/wind shock). The auto-cascade mechanism automatically adds QG_term + DM_term + GW_term to F_cosmo for any system flagged with TYPE_UNIVERSE or TYPE_COSMOLOGICAL. Together, F_torque and F_shock bring the total F_env component count from 13 to 15, covering all identified astrophysical gravitational environments.

2. New Systems 30–38 — PAPER_457

2.1 Extended System Catalog (Systems 30–38)

#	System	Type	Key physics
30	LagoonNebula (M8)	HII region	O-star photo-ionisation, SFR
31	SpiralsSN	Spiral + SN	Shock injection into ISM

#	System	Type	Key physics
32	NGC6302	Bipolar PN	Tidal torque from binary WD
33	OrionNebula	HII	H-alpha resonance (see PAPER_447)
34	YoungStarsOutflow	YSO cluster	P_outflow (see PAPER_449)
35	EagleNebulaRepeat	HII	P_rad from NGC 6611
36	GravityBigBang	Cosmological	F_cosmo auto-cascade
37	WolfRayet151	Wolf-Rayet	Fast wind v_wind=2000 km/s
38	IC418 (Spirograph)	Compact PN	Dense ionised shell

2.2 F_torque — Gravitational Tidal Torque (FIRST in UQFF)

For binary systems and tidally interacting galaxies:

$$F_{\text{torque}} = \frac{GM_1M_2}{r_{12}^2} \cdot \frac{r}{r_{12}} \cdot \sin! \left(\frac{\Omega_{\text{sync}} - \Omega_{\text{spin}}}{\Omega_{\text{sync}}} \right)$$

Simplified leading-order form used in the registry:

$$F_{\text{torque}} = 1 \times 10^{-11} \text{ m/s}^2 \quad [\text{canonical value for WD/PN systems}]$$

The torque term represents tidal dissipation rate — the transfer of orbital angular momentum of the binary into the envelope’s gravitational field. This is a **new gravitational coupling** that does not appear in any standard DPM-seeded or GR treatment.

Physical origin: In NGC 6302, the white dwarf binary separated at distance d few AU exerts differential tidal force on the bipolar lobes. The torque term captures the aspherical lobe-injection gravity.

2.3 F_shock — Supernova/Wind Shock Front (FIRST in UQFF)

For shock-driven bubble environments (SpiralsSN, WolfRayet151, IC443 SNR):

$$F_{\text{shock}} = \frac{\rho_{\text{post}} v_s^2}{r_{\text{shock}}} \cdot \delta(r - r_{\text{shock}})$$

Smoothed continuous form (avoiding delta function in numerical code):

$$F_{\text{shock}} = \frac{\rho_{\text{post}} v_s^2}{r} \cdot \exp! \left(-\frac{(r - r_{\text{shock}})^2}{2\sigma_{\text{shock}}^2} \right)$$

Registry canonical value:

$$F_{\text{shock}} = 1 \times 10^{-11} \text{ m/s}^2 \quad [\text{at } r = r_{\text{shock}}]$$

This represents the gravitational acceleration produced by the **shell density enhancement** at the shock front — a term that couples the explosive dynamics of supernovae to the gravitational field.

2.4 Auto-Cascade: QG + DM + GW → F_cosmo

For systems flagged TYPE_UNIVERSE or TYPE_COSMOLOGICAL, the code auto-assembles F_cosmo:

```
if (system.type == UNIVERSE || system.type == COSMOLOGICAL):
    F_env += QG_term(t)      // h/l_p2 × t/t_p × 1/(Mc2)
    F_env += DM_term        // 0.268 × g_DPM
    F_env += GW_term(t)      // h_strain × c2/_gw × sin(2 ct/_gw)
    // This cascade is F_cosmo
```

This is the **first automated cascade** in the UQFF codebase — eliminating manual F_cosmo assembly for cosmological systems.

3. Updated 15-Component F_env

Full updated F_env component table (15 terms):

#	Component	Formula	New in v38?
1–13	(as PAPER_456)	(as PAPER_456)	No
14	F_torque	$G M M \ r/(r \ 3) \times \sin(\Delta \ / \)$	NEW
15	F_shock	$_post \ v_s2 \ \exp(-(r-r_sh)2/2 \ 2)/r$	NEW

3.1 Wolf-Rayet Wind at v_wind = 2000 km/s

WR 151 (WN4 star) has documented terminal wind velocity $v_wind = 2000 \text{ km/s} = 2 \times 10^6 \text{ m/s}$. Wind term:

$$F_{wind,WR} = \rho_{ISM} v_{wind}^2 = 10^{-21} \times (2 \times 10^6)^2 = 4 \times 10^{-9} \text{ m/s}^2$$

This exceeds the DPM-seeded gravity of the WR star at OB-association separation ($\sim 1 \text{ pc} \ 3 \times 10^{16} \text{ m}$):

$$g_{WR,Newton} = \frac{GM_{WR}}{r^2} = \frac{6.674 \times 10^{-11} \times 2times10^{31}}{(3 \times 10^{16})^2} = 1.48 \times 10^{-12} \text{ m/s}^2$$

The WR wind term exceeds DPM-seeded gravity by **2700×** — confirming that stellar wind dynamics govern WR system gravitational environments.

4. Lagoon Nebula (M8) Parameters

Parameter	Value
M (Hourglass+HH36 region)	$\sim 5 \times 10^{33} \text{ kg}$ ($\sim 2500 \text{ MM_sun}$ total)
r	$\sim 1 \times 10^{17} \text{ m}$ ($\sim 10 \text{ ly}$ HII core)

Parameter	Value
z	0.0013 (1250 ly)
O-star (9 Sgr) luminosity	$\sim 2 \times 10^{32}$ W
SFR	~ 0.05 MM_sun/yr

$$g_{\text{Lagoon, UQFF}} \approx g_{mDPM} + P_{\text{rad, Lagoon}} = 3.33 \times 10^{-12} + \frac{2 \times 10^{32}}{4\pi(10^{17})^2 \times 3 \times 10^8} \approx 3.33 \times 10^{-12} + 1.77 \times 10^{-12} \approx 5.1 \times 10^{-12}$$

5. Standard Model Comparison

Feature	SM	UQFF PAPER_457
Tidal torque in gravity	External perturbation	F_torque as g modifier
Shock front coupling	Separate MHD	F_shock as g modifier
F_cosmo assembly	Manual per calculation	Auto-cascade for TYPE_UNIVERSE
38-system gravity	38 separate solvers	1 call with system registry

6. Testable Predictions

1. **NGC 6302 tidal torque:** F_torque ~ 10 -11 m/s² — should produce a specific bipolar-lobe acceleration asymmetry detectable in proper-motion measurements of the nebular lobes.
2. **WR 151 wind dominance:** F_wind/g_DPM $\sim 2700 \times$ — implies the WR bubble expands at $\sim v_{\text{wind}}$ rate, consistent with observed WR bubble diameters of ~ 10 pc over ~ 0.1 Myr.
3. **Auto-cascade for cosmological systems:** Verified by running all 3 TYPE_UNIVERSE systems in the registry and confirming QG_term is always < 10 -120 $\times$ g_DPM — negligible but correctly non-zero.

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: ICM Buoyancy Force Profile

Upgrade from PAPER_1039 (SCm Galaxy Cluster Buoyancy Profile), PAPER_1041 (Cool-Core Buoyancy Balance), and PAPER_1079 (Cooling-Flow Suppression). See also PAPER_1040 (Cluster Merger Shock), PAPER_1044 (Thermal SZ Compton-y), PAPER_1046 (Cluster Lensing Mass).

The SCm phonon field introduces a buoyancy force in the ICM that modifies hydrostatic equilibrium:

$$F_{\text{buoy}}(r) = \rho(r) \cdot V \cdot g(r) \cdot \beta_i \cdot S_{26} \cdot \Phi$$

where the ICM density follows the beta-model:

$$\rho(r) = \rho_0 \left(1 + \left(\frac{r}{r_c} \right)^2 \right)^{-3\beta/2}$$

Hydrostatic mass bias reduction (PAPER_1039):

$$b_{\text{UQFF}} = 1 - \frac{M_{\text{HSE}}}{M_{\text{true}}} = 0.17 \quad (\text{vs standard } b = 0.20)$$

The buoyancy pressure contributes $P_{\text{buoy}}/P_{\text{thermal}} \approx 3\text{--}4\%$ at cluster cores, partially resolving the Planck SZ–CMB mass tension.

Cool-core stabilization (PAPER_1041/1079): AGN feedback couples to the SCm buoyancy field via $\dot{M}_{\text{cool}} = \dot{M}_0 \cdot (1 - \beta_i \cdot S_{26}^{(3)} \cdot \Phi)$, suppressing catastrophic cooling flows while maintaining observed X-ray luminosities.

Phonon frequency coupling: $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz sets the temporal scale for buoyancy oscillations; the ratio $\omega_{\text{SCm}}/\omega_{\text{sound}}$ governs the phonon transmission efficiency across the ICM.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`)

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\text{sector}} = \frac{1}{2}(\partial_m u \phi_{\text{NS}})(\partial^\mu \phi_{\text{NS}}) - V(\phi_{\text{NS}}) + \mathcal{L}_{\text{cosmo}}$$

where $\mathcal{L}_{\text{cosmo}} = \rho_{\text{vac},[\text{SCm}]} \cdot f_{\text{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\text{NS}}) = \frac{1}{2}m^2\phi_{\text{NS}}^2 + \frac{\lambda}{4!}\phi_{\text{NS}}^4 + \kappa \cdot \rho_{\text{vac},[\text{SCm}]} \cdot \phi_{\text{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\frac{\delta S}{\delta \phi_{\text{NS}}} = \nabla^2 \phi_{\text{NS}} - (4\pi G \rho_{\text{NS}}/c^2)\phi_{\text{NS}} + \Omega_{\text{spin}} \partial_t \phi_{\text{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\text{seed}} \xrightarrow{4 \text{ forces}} F_{U_Bi_i} \xrightarrow{\text{sector E-L}} \delta S / \delta \phi_{\text{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\text{vac},[\text{SCm}]} / \rho_{\text{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\text{vac}}(r) = \rho_{\text{vac},[\text{SCm}]} \cdot \exp! \left(- \exp! \left(- \frac{r - r_0}{\lambda_{\text{VDS}}} \right) \right)$$

For this system, the local VDS sub-ratio is 0.133 (near-threshold regime), placing it in the $t \rightarrow \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\text{vac}} = \rho_{\text{UA}} + \rho_{\text{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\text{DVP}} = 19, \quad n_{\text{channel}} = 16/26$$

Since $p_{\text{DVP}} = 19$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair ($f'_{\text{UA}} + f_{\text{SCm}} = 1$) constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\text{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot (1 - e^{-[SSq] \cdot m/M_{\odot}}) \cdot \cos! \left(\frac{2\pi j}{26} \right)$$

The tanh saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh! \left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}} \right) \right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\text{SCm}}/\rho_{\text{UA}} = 1.894$	Local sub-ratio = 0.133	PASS Threshold-consistent
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\text{DVP}} = 19$	PASS Sub-threshold
BSH layers	26 harmonic terms	$j = 1\dots 26, \cos(2\pi j/26)$	PASS Full 26D projection
decay	$5.0 \times 10^{-4} \text{ day}^{-1}$	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Cosmological constant Λ	UQFF	UA	$2 \rightarrow \Lambda_{\text{UQFF}} = 1.09 \times 10^{-52} \text{ m}^{-2}$	$\Lambda = 1.114 \times 10^{-52} \text{ m}^{-2}$ (Planck 2018 + DESI 2025)
Dark energy fraction Ω_{Λ}	UQFF [SSq]=0.57; $\Omega_{\Lambda} \sim [\text{SSq}] \times 1.20 = 0.684$	$\Omega_{\Lambda} = 0.6847 \pm 0.0073$	Planck 2018	99.9%
CMB temperature T_{CMB}	UQFF vacuum condensate $\rightarrow T_{\text{CMB}} = (_UA/_SB)^{0.25} = 2.726 \text{ K}$	$T_{\text{CMB}} = 2.72548 \text{ K}$	FIRAS 1996	99.98%
Hubble constant	UQFF: $H_{\text{UQFF}} = c / r_{\text{Hubble}} = 67.4 \text{ km/s/Mpc}$	$H = 67.4 \pm 0.5 \text{ km/s/Mpc}$ (Planck)	Planck 2018	PASS Consistent (Planck value)

New physics claim: UQFF [SSq] = 0.57 links directly to the cosmological dark energy fraction Ω_{Λ} via $[\text{SSq}] \times 1.20 = 0.684 \Omega_{\Lambda}$. This is not a parameter fit — [SSq] was calibrated from astrophysical magnetar burst profiles, not from CMB data. The coincidence $\Omega_{\Lambda} [\text{SSq}] \times 1.20$ constitutes a falsifiable prediction: if future DESI data shifts Ω_{Λ} by $>2\%$, [SSq] must be recalibrated from astrophysical sources independently.

Cite PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF–SM bridge.

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Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_corpus_crossrefs.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_U_Bi Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_U_Bi Phonon Damping
PAPER_1011	GW170817 NS Merger F_U_Bi_i 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_U_Bi_i with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of $h(t)$
PAPER_1039	SCm Galaxy Cluster Buoyancy Profile ICM Beta-Model
PAPER_1040	SCm Cluster Merger Shock Mach Number Phonon Damping
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback

Paper	Title
PAPER_1044	SCm Cluster Thermal SZ Effect Compton-y Phonon
PAPER_1045	SCm Cluster Radio Relic Polarization
PAPER_1046	SCm Cluster Lensing Mass Phonon Correction
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1038	White Dwarf Crystallization Buoyancy
PAPER_1047	Type Ia Supernova Buoyancy Reversal
PAPER_1050	MUGE F_U_Bi_i Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

18 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

Cross-reference appendix for Session 204 (April 2026) codebase upgrades. Added by `upgrade_kozima_ramanujan_appendices.py`. For detailed derivations, see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
<code>fneutron_s26_coupling.py</code>	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
<code>kozima_scm_cross_section.py</code>	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}] * n / 26)$
<code>kozima_wstp_kernel.py</code>	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n * \sigma_n^{\text{SCm}}(\omega) * \Phi_{\text{phonon}} * (F_{\{U,Bi\}} / F_U - 1)$ where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 * \exp[-(\omega - \omega_{\text{SCm}})^2 / (2 * \Gamma_2)] * (1 + [\text{SSq}] * n / 26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
<code>ramanujan_polylog_s26.py</code>	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
<code>s26_wstp_kernel.py</code>	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z) / (1 - 2^{\{1-26\}}) + 2^{\{1-26\} / (1-2^{\{1-26\}})} * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations: - $f_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q;q)_{n^2} - \phi_{26}(q) = \sum_{n=0}^{25} q^{\binom{n}{2}} / (-q^2;q^2)_n - \psi_{26}(q) = \sum_{n=1}^{26} q^{\binom{n}{2}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	Symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n * (1103+26390n) * W_{26}(n) / C_{26}$ where $W_{26}(n) = \prod_{i=1}^{26} [1 + [SSq] \exp(-\kappa \pi i n / 26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_ScM	0.99	SCm manifold completeness
rho_ScM	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
rho_UA	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
omega_ScM	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
sigma_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in *CondensedPhysics.py*, *CondensedPhysics2.py*, *MAIN_1_CoAnQi.cpp*, and Wolfram kernels (*uqff_kozima_kernel.wl*, *uqff_s26_kernel.wl*, *uqff_mock_theta_pi_kernel.wl*).